

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 1

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 4 points below. 1 Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water 1 The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith 1 As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation 1 So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.2	1 mark each 1 mark for each of the 2 points below. 1 Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) 1 Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
01.3	1 mark each 1 mark for each of the 2 points below. 1 Flowing water carries solid particles – the ability to carry them depends on the water velocity and the density of the solids 1 As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Limestone / chalk from the shells of marine organisms 1 Coal from terrestrial vegetation 1 Crude oil and natural gas from marine organisms (plankton) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
02.1	1 mark for the point below. 1 As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.2	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $2400 / 40 = 60$ times Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (times) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	1

Question	Marking guidance	Marks
02.3	1 mark each 1 mark for each of the 3 points below. 1 Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) 1 Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits 1 The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round.	3
02.4	1 mark each 1 mark for each of the 2 points below. 1 Lowering the cut-off ore grade includes deposits of lower purity in the reserves 1 Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	1 mark each 1 mark for each of the 3 points below. 1 The total quantity (stock / resource base) is far greater than could ever be used 1 But most is at purities too low or in locations too inaccessible for current economic exploitation 1 The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable – energy needs rise rapidly as ore grade falls Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	

Question	Marking guidance	Marks
03.1	1 mark each 1 mark for each of the 4 points below. 1 Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) 1 Magnetometry 1 Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata 1 Resistivity Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.2	1 mark each 1 mark for each of the 3 points below. 1 Trial drilling is the most expensive technique per site 1 Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites 1 Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. remote sensing → collecting data from aircraft or satellites	3
03.3	1 mark each 1 mark for each of the 2 points below. 1 Different minerals emit / reflect infrared at different wavelengths 1 Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth	2
03.4	1 mark for the point below. 1 Results are available immediately in the field, saving time waiting for laboratory analysis Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	

Question	Marking guidance	Marks
04.1	1 mark each 1 mark for each of the 2 points below. 1 The lowest ore purity that can be mined economically with existing technology 1 Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	1 mark each 1 mark for each of the 6 points below. 1 X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper 1 But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) 1 X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs 1 X is far from transport infrastructure so transport costs are high 1 Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall 1 Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching) – both sulfides risk acid mine drainage Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.3	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Higher market value / price of the metal 1 Improved extraction technology 1 Lower energy costs Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
05.1	1 mark each 1 mark for each of the 2 points below. 1 Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports 1 Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones – open-cast mines take far more land than deep mines Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
05.2	1 mark each 1 mark for each of the 5 points below. 1 Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle 1 Noise: baffle mounds / embankments absorb and deflect – blasting at set, predictable daytime hours – quieter machinery 1 Turbid water: sedimentation lagoons where water stands until solids settle before discharge 1 Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals 1 Amenity: landscaping and tree planting to screen the mine – restoration plans agreed as a condition of planning permission Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Leachate → polluted liquid draining out of waste	5
05.3	1 mark each 1 mark for each of the 3 points below. 1 Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface 1 At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village 1 Killing 144 people, mainly children in the primary school Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • waterlogging → soil pores filled with water so there is no air	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
06.1	1 mark each 1 mark for each of the 2 points below. 1 The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions 1 The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	1 mark each 1 mark for each of the 2 points below. 1 Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) 1 Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
06.3	1 mark for the point below. 1 Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
07.1	1 mark each 1 mark for each of the 5 points below. 1 Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions 1 Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected – recirculation increases the metal concentration 1 Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) 1 Smelting needs chemical reduction at high temperatures – energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic 1 Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. sustainable → meeting present needs without stopping future generations meeting theirs Leachate → polluted liquid draining out of waste Do not credit 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
07.2	1 mark each 1 mark for each of the 3 points below. 1 Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves 1 The vegetation is harvested and incinerated 1 The concentrated metal in the ash is dissolved in acid and separated by electrolysis OR decontaminates polluted sites Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Question	Marking guidance	Marks
07.3	1 mark each 1 mark for each of the 2 points below. 1 Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) 1 The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. adsorption → nutrient ions sticking to the surfaces of clay and humus	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
08.1	1 mark each 1 mark for each of the 3 points below. 1 Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium 1 Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet 1 International agreement on ownership of the seabed and its resources is needed Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.2	1 mark each 1 mark for each of the 3 points below. 1 Collectors disturb the seabed and kill benthic organisms living on it 1 Separating nodules from sediment creates plumes that increase turbidity with unknown consequences 1 Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Question	Marking guidance	Marks
08.3	1 mark each 1 mark for each of the 4 points below. 1 Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) 1 Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste 1 Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them 1 Deep sea sediments contain relatively high concentrations but no viable extraction method yet Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. adsorption → nutrient ions sticking to the surfaces of clay and humus Do not credit 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	